$$\hat{p} = \frac{\hbar}{i}\nabla = \frac{\hbar}{i}\left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3}\right)$$

$$\vec{x} = (x_1, x_2, x_3)$$

$$[\hat{x}_i, \hat{p}_i] = i\hbar \qquad [\hat{x}_i, \hat{p}_j] = i\hbar\,\delta_{ij}$$

Sch. Eq^n. in 3D →

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\vec{x})\psi(\vec{x}) = E\psi(\vec{x})$$

↳ Time Independent.

$$V = V(|\vec{x}|) = V(r) \quad ; r = |\vec{x}|$$

→ Spherically symmetric potential
invariant under rotation
→ Central potential.

$$\psi = \psi(r, \theta, \phi)$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right) + \underbrace{V(r)}_{\text{Central Potential.}}\psi = E\psi$$

* **Orbital Angular momentum** →

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$$
$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$$
$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

$$[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z]$$
$$= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z]$$

$$[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}$$

$$[\hat{L}_x \ \hat{L}_y] = \cancel{[\hat{y}\hat{p}_z, \hat{p}_x\hat{z}]\hat{p}_x} + [redacted]$$
$$= [\hat{y}\hat{p}_z, \hat{z}]\hat{p}_x + \hat{x}[\hat{z}\hat{p}_y, \hat{p}_z]$$
$$\cancel{+ \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x +}$$
$$= \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y$$
$$= \hat{y}(-i\hbar)\hat{p}_x + \hat{x}(i\hbar)\hat{p}_y$$
$$= i\hbar[\hat{x}\hat{p}_y - \hat{y}\hat{p}_x] = i\hbar\hat{L}_z$$

$$\underbrace{\begin{aligned}[\hat{L}_x, \hat{L}_y] &= i\hbar \hat{L}_z \\ [\hat{L}_y, \hat{L}_z] &= i\hbar \hat{L}_x \\ [\hat{L}_z, \hat{L}_x] &= i\hbar L_y\end{aligned}}$$

$\hookrightarrow$ Do not commute with eachother
No simultaneous eigenstates.

$$* \ \hat{L}^2 = \hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z$$

$$[L_z, \hat{L}^2] = [\hat{L}_z, \hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z]$$
$$= [\hat{L}_z, \hat{L}_x\hat{L}_x] + [\hat{L}_z, \hat{L}_y\hat{L}_y]$$
$$= [\hat{L}_z, \hat{L}_x]\hat{L}_x + \hat{L}_x[\hat{L}_z, \hat{L}_x] + [\hat{L}_z, \hat{L}_y]\hat{L}_y + \hat{L}_y[\hat{L}_z, \hat{L}_y]$$
$$= \cancel{i\hbar\hat{L}_y\hat{L}_x} +$$

$$= \cancel{i\hbar\hat{L}_y\hat{L}_x} + \cancel{\hat{L}_x i\hbar \hat{L}_y} - \cancel{i\hbar\hat{L}_x\hat{L}_y} - \cancel{\hat{L}_y i\hbar \hat{L}_x}$$

$$= 0$$

$\Rightarrow \hat{L}_z, \hat{L}^2$ have common eigenstates.

$$x = r\sin\theta\cos\phi \qquad r = \sqrt{x^2+y^2+z^2}$$
$$y = r\sin\theta\sin\phi \qquad \theta = \cos^{-1}(z/r)$$
$$z = r\cos\theta \qquad \phi = \tan^{-1}(y/x)$$

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \cancel{\frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}}^{\,0}$$
$$= -r\sin\theta\sin\phi\frac{\partial}{\partial x} + r\sin\theta\cos\theta\frac{\partial}{\partial y}$$
$$= -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$$

$$\frac{\hbar}{i}\frac{\partial}{\partial\phi} = x\hat{p}_y - y\hat{p}_x = \hat{L}_z$$

$$\Rightarrow \boxed{\hat{L}_z = \frac{\hbar}{i}\frac{\partial}{\partial\phi}}$$

$$\boxed{\begin{aligned}\hat{L}_x &= -\frac{\hbar}{i}\left(\sin\theta\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)\\ \hat{L}_y &= \frac{\hbar}{i}\left(\cos\theta\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)\end{aligned}}$$

$$\hat{L}^2 = -\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)$$

Let $\Psi_{\ell m}(\theta,\phi)$ are the simultaneous eigenstates of $\hat{L}_z$, $\hat{L}^2$

$$\hat{L}_z\Psi_{\ell,m}(\theta,\phi) = m\hbar\,\Psi_{\ell,m}(\theta,\phi)$$
$$\hat{L}_z f(\phi) = \lambda f(\phi)$$

$$\Rightarrow \frac{df}{d\phi} - \frac{i\lambda}{\hbar}f = 0$$

$$\Rightarrow \frac{df}{d\phi} \quad f(\phi) = c e^{(i\lambda/\hbar)\phi}$$

$$f(\phi + 2\pi) = f(\phi)$$

$$e^{\frac{i\lambda}{\hbar}\phi} e^{\frac{i\lambda}{\hbar}\cdot 2\pi} = e^{\frac{i\lambda}{\hbar}\phi} = \cancel{f(\phi)}$$

$$e^{\frac{2\pi i\lambda}{\hbar}} = 1$$

$$\Rightarrow \lambda = m\hbar \quad ; \quad m = 0, \pm 1, \pm 2 \ldots$$

$$\frac{\partial \Psi_{\ell m}}{\partial \phi} = im \Psi_{\ell m} \Rightarrow \Psi_{\ell m}(\theta, \phi) = e^{im\phi} P_\ell^m(\theta)$$

Eigenvalues of $\hat{L}_z$: $m\hbar$, $m = 0, \pm 1, \pm 2 \ldots$

Eigenfx^n s. of $\hat{L}_z$: $c e^{im\phi} = f(\phi)$

$$\int_0^{2\pi} |f(\phi)|^2 d\phi = 1$$

$$\Rightarrow c = \frac{1}{\sqrt{2\pi}}$$

We claim that $\hat{L}^2 Y_{\ell m}(\theta, \phi) = \ell(\ell+1)\hbar^2 Y_{\ell,m}(\theta, \phi)$

$$-\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \cancel{\cot} \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right) Y_{\ell,m} = \ell(\ell+1)\hbar^2 Y_{\ell m}$$

$$\left[\sin\theta \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{\partial^2}{\partial\phi^2} \right] Y_{\ell,m} = -\ell(\ell+1) Y_{\ell m} \sin^2\theta$$

$$Y_{\ell m}(\theta, \phi) = e^{im\phi} P_\ell^m(\theta)$$

$$\frac{\partial^2}{\partial\phi^2} (e^{im\phi}) = -m^2 e^{im\phi}$$

$$\sin\theta \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial P_\ell^m(\theta)}{\partial\theta} \right) - m^2 P_\ell^m(\theta) = -\ell(\ell+1)\sin^2\theta P_\ell^m(\theta)$$

$$\sin\theta \frac{d}{d\theta} \left(\sin\theta \frac{d P_\ell^m(\theta)}{d\theta} \right) + \left[\ell(\ell+1)\sin^2\theta - m^2 \right] P_\ell^m(\theta) = 0$$

$$x = \cos\theta \quad \Rightarrow \frac{d}{d\theta} = \frac{dx}{d\theta}\cdot\frac{d}{dx} = -\sin\theta\frac{d}{dx}$$

$$\sin\theta\frac{d}{d\theta} = -\sin^2\theta\frac{d}{dx} = -(1-x^2)\frac{d}{dx}$$

$$\Rightarrow \cancel{(1-x^2)}\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}P_l^m(x)\right] + \left[l(l+1)\cancel{(1-x^2)} - \frac{m^2}{(1-x^2)}\right]P_l^m(x)$$

$P_l^m(x)$: Associated Legendre fx^n.

$m=0$: $P_l^0(x) = P_l(x) \rightarrow$ Legendre Polynomial

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_l(x)}{dx}\right] + l(l+1)P_l(x) = 0$$

$$P_0(x) = 1 \; ; \; P_1(x) = x \; , \; P_2(x) = \frac{1}{2}(3x^2-1)$$

Take soln. of $P_2(x)$

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_2(x)}{dx}\right] + \lambda P_2(x) = 0$$

$$\frac{d}{dx}\left[(1-x^2)(3x)\right] + \frac{\lambda}{2}(3x^2-1) = 0$$

$$3 - 9x^2 + \frac{\lambda}{2}(3x^2-1) = 0$$

$$\frac{\lambda}{2}\times 3 = 9 \Rightarrow \lambda = 6$$

$$6 = 2(2+1) \equiv l(l+1)$$

Possible values of l: 0, 1, 2 ....

Possible values of m for a given l: $-l \leq m \leq +l$

$$\hat{L}^2 Y_{lm}(\theta,\phi) = l(l+1)\hbar^2 Y_{lm}(\theta,\phi)$$

$Y_{lm}(\theta,\phi) \longrightarrow$ Spherical harmonics $= P_l^m(\theta)e^{im\phi}$

$L_z Y_{lm}(\theta,\phi) \rightarrow m\hbar Y_{lm}(\theta,\phi)$

$\psi(L) = 0$

$\psi(x) = \psi(x + L)$

- $\hat{L}_z$, $\hat{L}^2 \rightarrow$ Involve derivatives wrt angular co-ordinates only. $\Rightarrow [\hat{L}_z, f(r)] = 0$

$[\hat{L}_z, \hat{L}^2] = 0$

$$\hat{H} = \frac{-\hbar^2}{2m}\frac{1}{\partial r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2mr^2} + V(r)$$

$[\hat{L}_i, \hat{H}] = 0$, $[\hat{L}^2, \hat{H}] = 0$

$\hat{H}$, $\underbrace{\hat{L}^2, \hat{L}_i}_{\text{constants of motion (good quantum numbers)}}$ $\rightarrow$ common eigen fn.

$\rightarrow$ commute with each other.

$$\hat{H} = \frac{-\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)\right] + V(r)$$

$-\hbar^2(\;) = \hat{L}^2$

$$\hat{H}\psi = \frac{-\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{\hat{L}^2}{2mr^2}\psi + V(r)\psi = E\psi$$

$$\psi = R(r)\,Y_{lm}(\theta,\phi)$$

$$\frac{-\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)Y + \frac{\hat{L}^2}{2mr^2}RY + V(r)RY = ERY$$

Multiply by $\frac{r^2}{RY}$

$$\frac{-\hbar^2}{2m}\cancel{\frac{1}{r^2}}\frac{1}{R}\frac{d}{dr}\left(\cancel{R}r^2\frac{dR}{dr}\right) + \frac{1}{2mY}\hat{L}^2Y + V(r)r^2 = Er^2$$

$$\frac{-\hbar^2}{2m}\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + (V(r) - E)r^2 = \frac{-1}{2mY}\hat{L}^2Y = \lambda$$

$$\lambda = \frac{-l(l+1)\hbar^2}{2m}$$

$$\frac{-\hbar^2}{2m}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + [V(r) - E]r^2R = -\frac{l(l+1)\hbar^2}{2m}R$$

$$\frac{-\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{l(l+1)\hbar^2}{2mr^2} + V(r)\right]R = ER$$

We can reduce this eq^n. in 1D with $V_{eff}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}$

General soln. :- $u_l(r) = \left(\sum_{i=1}^{n} a_i r^i\right) e^{-\beta r}$

$l=0$: $\frac{-\hbar^2}{2m}\frac{d^2u}{dr^2} - \frac{Ze^2}{r}u = Eu$

$n=1$: $u = re^{-\beta r}$; $R = \frac{u}{r} = e^{-\beta r}$

$\Rightarrow \left(\frac{\hbar^2\beta}{m} - Ze^2\right) - \frac{\hbar^2\beta^2}{2m} r = Er$

$E = -\frac{\hbar^2\beta^2}{2m}$; $Ze^2 = \frac{\hbar^2\beta}{m}$

$\Downarrow$

$\beta^2 = \frac{2m|E|}{\hbar^2}$

$\Rightarrow \frac{\hbar^2}{m}\sqrt{\frac{2m|E|}{\hbar^2}} = Ze^2$

$\boxed{|E| = \frac{mZ^2e^4}{2\hbar^2}} \longrightarrow$ Ground State Energy of H-atom.

* Bohr Radius : $a_0 = \frac{\hbar^2}{me^2} = 0.529 Å$

$|E| = \frac{e^2Z^2}{2a_0}$; $\frac{e^2}{2a_0} = \frac{|e|^4}{2}\left(\frac{m}{\hbar^2}\right)$

$= \frac{1}{2}\left(\frac{e^4}{\hbar^2c^2}\right) \underbrace{mc^2}_{\text{Rest mass energy of } e^- = 0.511 \text{ MeV}}$

Fine structure constant : $\alpha = \frac{e^2}{\hbar c} = \frac{1}{137}$

$E = -|E| = -13.6 Z^2$ eV

For H : $Z = 1 \Rightarrow E = -13.6$ eV

Ground State : $n=1$, $l=0$ $\quad u = C_N r e^{-\beta r}$

$R = \frac{u}{r} = C_N e^{-\beta r}$

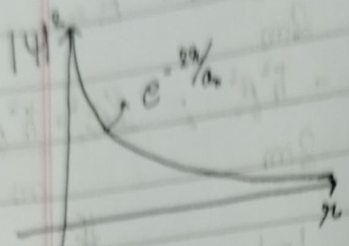
Ψ form : $\Psi_{100} = C_N e^{-\beta r} Y_{lm}(\theta,\phi)$

$\beta = \sqrt{\frac{2m|E|}{\hbar^2}} = \sqrt{\frac{me^2Z^2}{a_0\hbar^2}} = \frac{1}{a_0}$

Normalize → $4\pi|C_1|^2 \int_0^\infty r^2 e^{-\frac{2r}{a_0}} dr = 1$

$\Rightarrow C_1 = \frac{1}{\sqrt{\pi a_0^3}}$

H-atom ground state: $\boxed{\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-\frac{r}{a_0}}}$



* A more rigorous approach → Radial Part

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} - \frac{Ze^2}{r}\right)u = Eu$$

$K^2 = -\frac{E}{2Z^2e^2/a_0}$ $r = \frac{a_0}{2Z}x$; $\rho = Kx = \frac{2KZ}{a_0}r$

$$\left[-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{1}{K\rho}\right]u = -u$$

$\rho \to 0 \Rightarrow u \to 0$ & $\frac{l(l+1)}{\rho^2}$ dominates over $\frac{1}{K\rho}$

$$\frac{d^2u}{d\rho^2} - \frac{l(l+1)}{\rho^2}u = 0$$

$\Rightarrow$

$$u = \rho^{l+1}$$

$\rho \to \infty \Rightarrow u = e^{-\rho}$

General Soln :- $u(\rho) = \rho^{l+1} \underbrace{w(\rho)}_{\text{Polynomial}} e^{-\rho}$

$$\rho \frac{d^2 w}{d\rho^2} + 2(l+1-\rho)\frac{dw}{d\rho} + \left[\frac{1}{K} - 2(l+1)\right] w = 0$$

$$w = \sum_{k=0}^{\infty} a_k \rho^k$$

$$\frac{dw}{d\rho} = \sum_{k=0}^{\infty} a_k k \rho^{k-1} \; ; \; \frac{d^2 w}{d\rho^2} = \sum_{k=0}^{\infty} a_k k(k-1) \rho^{k-2}$$

Plug in the expressions & collect coeffs. of ρ^k

$$a_{k+1}(k+1)k + 2(l+1) a_{k+1}(k+1) - 2a_k k + \left[\frac{1}{K} - 2(l+1)\right]$$

$$\Rightarrow \frac{a_{k+1}}{a_k} = \frac{\left[2(k+l+1) - \frac{1}{K}\right]}{(k+1)(k+2l+2)}$$

In the large k limit $\Rightarrow \frac{a_{k+1}}{a_k} \cong \frac{2}{k+1}$

$k=0$: $a_1 = 2a_0 = \frac{2^1}{1!} a_0$

$k=1$: $a_2 = \cancel{2a_1} = \frac{2^2}{2!} a_0$

$k=2$: $a_3 = \frac{2^3}{3!} a_0$

$$w = \sum_{k=0}^{\infty} a_k \rho^k = \sum_{k=0}^{\infty} \frac{2^k \rho^k a_0}{k!} = \sum_{k=0}^{\infty} \frac{(2\rho)^k}{k!} a_0$$

$$= e^{2\rho} a_0$$

$\therefore \sim \rho^{l+1} w e^{-\rho} \Rightarrow w$ becomes unnormalizable.

So, w needs to be truncated.

Suppose w is a polynomial of degree N

$a_N \neq 0$, $a_{N+1} = 0 \Rightarrow a_{N+2}\ a_{N+3} \dots = 0$

$$\Rightarrow \frac{1}{K} = 2(N + l + 1)$$

↳ Contains information about E

⇒ E is quantised.

For each combination of N & l → expression for E

Principal quantum number : $\boxed{n = N + l + 1} = \dfrac{1}{2K}$

$$K^2 = \frac{-E}{(2Z^2e^2/a_0)}$$

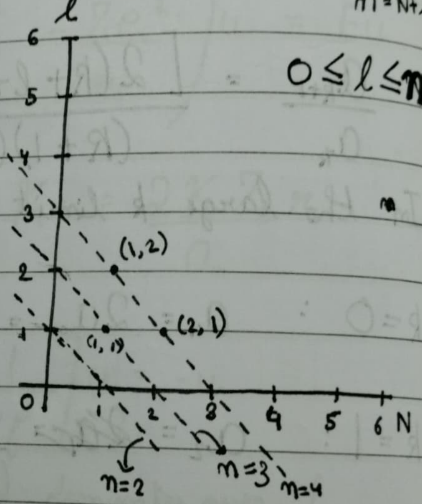
$$E = -\frac{2Z^2e^2}{a_0} K^2$$

$$= -\frac{2Z^2e^2}{a_0^2} \cdot \frac{1}{4K^2}$$

$$\boxed{E_n = -\frac{Z^2e^2}{2a_0 n^2}}$$

$n = N + l + 1$

$0 \leq l \leq n-1$



G.S. : $E_1 = -\dfrac{\cancel{Z}e^2}{2a_0}$

for H

$n = 1$, $l = 0$ → 1 state

$n = 2$, $l = 1, 0$ ⟶ 4 states

($l = 1$ ↓ $m = -1, 0, 1$ 3 states)

$n = 3$, $l = 2, 1, 0$ ⟶ 9 states

In general for n ⟶ n^2 states.

Quantum numbers: n, l, m

Once we fix $n, l,$ — N is automatically fixed.

$$\rho = \frac{2KZr}{a_0} = \frac{Zr}{na_0}$$

$$\psi_{nlm} \sim \rho^{l}\, W_{nl}(\rho)\, e^{-\rho}\, Y_{lm}(\theta,\phi)$$

$$\boxed{\psi_{nlm}(r,\theta,\phi) = A\left(\frac{r}{a_0}\right)^{l} W_{nl}\left(\frac{r}{a_0}\right) e^{-\frac{Zr}{na_0}}\, Y_{lm}(\theta,\phi)}$$

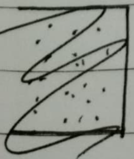
GS:- $\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}}\, e^{-r/a_0}$ $(Z=1)$

	$l=0$	$l=1$	$l=2$	$l=3$
$-\frac{1}{16}$	$N=3$	$N=2$	$N=1$	$N=0$
$-\frac{1}{9}$	$N=2$	$N=1$	$N=0$	
$-\frac{1}{4}$	$N=1$	$N=0$		
-1	$N=0$			
n				

$Z \neq 1$: $V(r) = \frac{e^2}{r} \longrightarrow \frac{Ze^2}{r}$

$$a_0 = \frac{\hbar^2}{me^2} \longrightarrow \frac{\hbar^2}{Zme^2} = \frac{a_0}{Z}$$

$$\boxed{\psi_{100} = \sqrt{\frac{Z^3}{\pi a_0^3}}\, e^{-\frac{rZ}{a_0}}}$$



2- $\Psi(\vec{r}) = c\left[\phi_{100}(\vec{r}) + 4i\,\phi_{210}(\vec{r}) - 2\sqrt{2}\,\phi_{21-1}(\vec{r})\right]$

$\hat{H}\phi_{n\ell m} = E_n \phi_{n\ell m}$

$\hat{L}^2 \phi_{n\ell m} = \ell(\ell+1)\hbar^2 \phi_{n\ell m}$

$\hat{L}_z \phi_{n\ell m} = m\hbar\, \phi_{n\ell m}$

$\Psi^*(\vec{r}) = c\left[\phi^*_{100}(\vec{r}) - 4i\,\phi^*_{210}(\vec{r}) - 2\sqrt{2}\,\phi^*_{21-1}(\vec{r})\right]$

$\int \Psi^* \Psi \, d^3 r = 1$ $\Rightarrow$ Cross terms vanish

$|c|^2 \left[1 + 16 + 8\right] = 1$

$|c|^2 = 1/25 \Rightarrow c = 1/5$

$\langle \hat{H} \rangle = |c|^2 \left[E_1 + 16 E_2 + 8 E_2\right]$

$= \frac{1}{25}(E_1 + 24 E_2)$

$\langle \hat{L}^2 \rangle = |c|^2 \left[0 + 16 \times 2\hbar^2 + 8 \times 2\hbar^2\right]$

$= \frac{48}{25}\hbar^2$

$\langle \hat{L}_z \rangle = |c|^2 \left[0 + 0 + 8(-1)\hbar\right]$

$= -\frac{8}{25}\hbar$

$\Psi(\vec{r}, t) = c\left[\phi_{100}\, e^{-\frac{iE_1 t}{\hbar}} + 4i\,\phi_{210}\, e^{-\frac{iE_2 t}{\hbar}} - 2\sqrt{2}\,\phi_{21-1}\, e^{-\frac{iE_2 t}{\hbar}}\right.$

3- $\Psi_{nlm} = \left(\frac{r}{a_0}\right)^l w_{nl}\left(\frac{r}{a_0}\right) e^{-\frac{Zr}{a_0}} Y_{lm}(\theta,\phi)$

$l = n-1$; $N = n - (\underbrace{l}_{n-1} + 1) = 0$

$\langle r \rangle = A^2 \int \psi^*_{n,n-1,m}\, r\, \psi_{n,n-1,m}\, r^2\, d\Omega$

$= A^2 \int r^3 \left(\frac{2r}{na_0}\right)^{2n-2} e^{-\frac{2r}{na_0}} dr \cancel{\int Y^*_{n-1,m} Y_{n-1,m} d\Omega}$ ①

$A^2 = \left(\frac{2}{na_0}\right)^3 \frac{1}{2n(2n-1)!}$

$\langle r \rangle = \left(\frac{2}{na_0}\right)^3 \frac{1}{2n(2n-1)!} \int r^3 \left(\frac{2r}{na_0}\right)^{2n-2} e^{-\frac{2r}{na_0}} dr$

$= \frac{1}{2n(2n-1)!} \int \left(\frac{2r}{na_0}\right)^{2n+1} e^{-\frac{2r}{na_0}} dr$

$= \frac{na_0}{2} \cdot \frac{1}{2n(2n-1)!} \int_0^\infty z^{2n+1} e^{-z} dz$

$= \frac{na_0}{2} \cdot \frac{1}{2n(2n-1)!} \times (2n+1)!$

$= \frac{na_0}{2}(2n+1) = n^2 a_0 \left(1 + \frac{1}{2n}\right) \approx n^2 a_0$

$\langle r^2 \rangle = (na_0)^2 \left(n + \frac{1}{2}\right)(n+1)$

$\sqrt{\langle r^2 \rangle} = \sqrt{n^4 a_0^2 \left(1+\frac{1}{2n}\right)\left(1+\frac{1}{n}\right)} \simeq n^2 a_0$

$\Delta r = \sqrt{\langle r^2 \rangle - \langle r \rangle^2}$

$= \sqrt{n^2 a_0^2 \left(n+\frac{1}{2}\right)\cancel{(n+1)} - \cancel{n^4 a_0^2}}$

$= na_0 \sqrt{\left(n+\frac{1}{2}\right)\frac{1}{2}}$

$\frac{\Delta r}{\langle r \rangle} \to 0 \Rightarrow e^-$ is localized around r

$$\frac{\Delta r}{\langle r \rangle} = \frac{na_0 \sqrt{(n+\frac{1}{2})\frac{1}{2}}}{\frac{na_0}{2}(2n+\frac{1}{2})} = \frac{1}{\sqrt{2(n+\frac{1}{2})}} \to 0 \text{ for large } n$$

4- $r \to \infty$, $\frac{-\hbar^2}{2m}\frac{d^2u}{dr^2} = Eu$

$$E_n = -\frac{Z^2e^2}{2a_0}\frac{1}{n^2}$$

$$= -\frac{Z^2}{2a_0} \times \frac{\hbar^2}{ma_0} \times \frac{1}{n^2}$$

$$\Rightarrow \frac{-2mE_n}{\hbar^2} = \frac{Z^2}{n^2a_0^2}$$

$$\frac{d^2u}{dr^2} = \frac{Z^2}{n^2a_0^2}u$$

$$\Rightarrow u = e^{-\frac{Zr}{na_0}}$$

5- $\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$

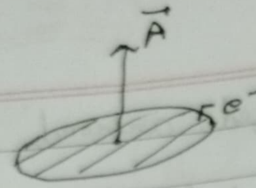
$$\langle r \rangle = \frac{1}{\pi a_0^3}\int_0^\infty r e^{-\frac{2r}{a_0}} \cdot 4\pi r^2 dr$$

$$= \frac{3a_0}{2}$$

$\langle r^2 \rangle = 3a_0^2$

By symmetry, $\langle x^2 \rangle = \frac{\langle r^2 \rangle}{3} = a_0^2$

$\langle x \rangle = 0$ [equal prob. of $\oplus$ & $\ominus$]

* 

$$I = \frac{e}{T} = \frac{ev}{2\pi R}$$

$$|\vec{\mu}| = I|\vec{A}| = \frac{ev}{2\pi R} \cdot \pi R^2$$

$$= \frac{1}{2} evR = \frac{e}{2m}(mvR)$$

$$= \frac{e}{2m}|\vec{L}|$$

$$\vec{\mu} = -\frac{e}{2m}\vec{L}$$

For intrinsic spin angular momentum, $\vec{S}$

$$|\vec{\mu}| = g\frac{e}{2m}|\vec{S}|$$

$g = 2$ for electron

$$= g\left(\frac{e\hbar}{2m}\right)\frac{|\vec{S}|}{\hbar}$$

$\frac{e\hbar}{2m} \rightarrow$ Bohr Magneton : μ_B
$= 9.27 \times 10^{-24}$

$$\boxed{\vec{\mu} = -g\mu_B\frac{\vec{S}}{\hbar}}$$

Ag —

* Stern-

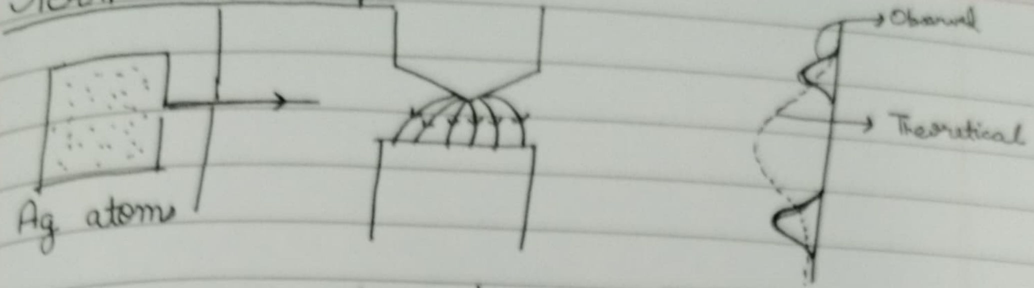
Ag a

*

Ag ↳ l is $5s^1$: $l=0$

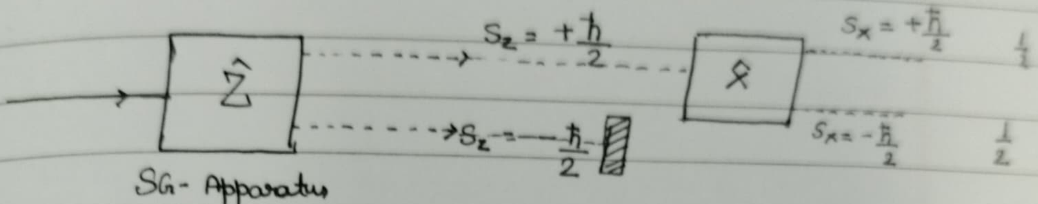
∴ Ag doesnot have any orbital angular momentum.

* Stern-Gerlach expt

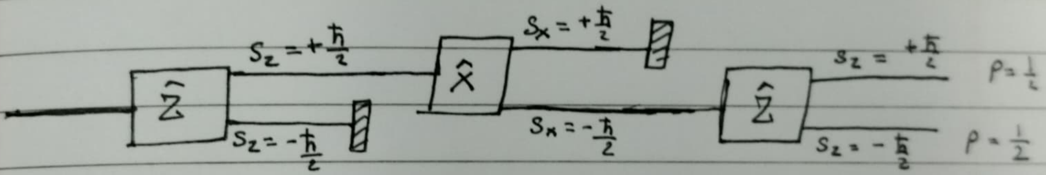


$$\vec{F} = \vec{\nabla}\cdot(\vec{\mu}\cdot\vec{B}) = \mu_z \frac{dB}{dz}\hat{z}$$

$$\boxed{\frac{S_z}{\hbar} = \pm\frac{1}{2}}$$



$$\hat{S}_z |z;+\rangle = +\frac{\hbar}{2}|z;+\rangle$$
$$\hat{S}_z |z;-\rangle = -\frac{\hbar}{2}|z;-\rangle$$



* Inner Product : * $(\phi, \psi) = \int dx\, \phi^*(x)\,\psi(x)$

* $(\psi_1, \psi_2) = \langle\psi_1|\psi_2\rangle$

General 2x2 ~~real~~ hermitian matrix

$$\hat{H} = \begin{pmatrix} a_0 + a_3 & a_1 - ia_2 \\ a_1 + ia_2 & a_0 - a_3 \end{pmatrix} = (H^T)^*$$

$$\hat{H} = a_0\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_1\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + a_2\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + a_3\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\hat{H} = a_0\sigma_0 + a_1\sigma_1 + a_2\sigma_2 + a_3\sigma_3$$
$$= \sum_{i=1}^{4} a_i \sigma_i$$

σ_i ↳ Pauli Matrices.

$$\hat{S}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$